SHIHAO ZHUO

(+86) 17858885610 · zhuoshihao99@gmail.com · cv-zhuo.github.io

EDUCATION

University of Birmingham Birmingham, UK
MSc, Artificial Intelligence and Machine Learning 2026.09 - present

Ningbo University Ningbo, China
BEng, Computer Science and Technology 2018.09 - 2024.06

- The six-year span includes two years of compulsory military service (2019.09–2021.09).

EXPERIENCE

HKUST (Guangzhou), Research Assistant, Data-Centric AI 2026.03-2026.07

- Method design, experiments and paper writing for the RMD^2 framework, targeting PVLDB (see Publications and Patents).
- Key design: replacing the absolute Mahalanobis distance with an inter-class margin as the label-consistency criterion, so that hard but correctly labelled samples are no longer discarded; ablations identify this as the main source of detection performance.
- Built a two-modality pipeline (DINOv2-ViT-S/14, BGE-large-en-v1.5, ResNet-9, BERT-base, Qwen2.5-1.5B with LoRA/DPO) on 4×RTX A5000, benchmarked under identical data splits against both proxy-based data-attribution methods and proxy-free feature-space selectors.

Shanghai Jiao Tong University, Research Assistant 2024.06-2024.12

- Independently built and maintained the research group website on XAMPP 8.0 and WordPress 5.8 (`https://aipsybrain.com/`).
- Independently developed a Python GUI application for cognitive assessment tasks, supporting memory and cognition scales; deployed for hospital-side data collection.

Anlai Scientific Instruments (Ningbo) Co., Ltd., Intern Assistant 2024.07-2024.10

- Applied YOLOv8 to a mouse scratching-behaviour analysis project: detection data processing and model tuning.
- Audited annotation quality and tuned algorithm parameters to keep training data accurate and model performance stable.

Ningbo Fanxing Consulting Co., Ltd., Development Engineer (Intern) 2023.04-2023.07

- Built a college-application recommendation system as a WeChat Mini Program on WeChat Cloud's database; the work obtained one software copyright.

RESEARCH PROJECTS

- **Multi-Center MRI Image Classification Based on Deep Learning** 2024.01-2024.05
  To address the loss of generalisation caused by multi-center data heterogeneity, diffusion models and domain-transfer techniques align unseen-center data with known-center data, improving cross-center transferability of the classifier; the method was filed as a patent.
- **Accurate Rib Fracture Diagnosis with Lightweight Deep Models** 2022.03-2023.03
  A lightweight network for chest CT that localises and classifies rib fracture regions and generates diagnostic reports, raising an alert when the radiologist's reading disagrees with the model prediction, so as to reduce missed and incorrect diagnoses.

PUBLICATIONS AND PATENTS

- **RMD^2: One Geometry, Two Roles in Data Curation and Generation Guidance** **In preparation, PVLDB**
  Co-first author. A low-cost curation operator for large data pipelines: class-conditional statistics are fitted once on a small trusted reference set, after which scoring a candidate is one encoder pass and one quadratic form, with no task-proxy model trained anywhere in the loop. The contribution is to judge label consistency by an inter-class margin rather than an absolute distance, and to make the same fitted statistics serve both noisy-sample curation and generation guidance.

- **Multi-Center MRI Image Classification Method**, invention patent 2024.08
  Granted, CN117173489B; **first inventor**. A vertical loss separates the content features of an image from its style features, so that the classifier attends only to content. This mitigates heterogeneity between imaging centers and strengthens cross-center generalisation.
- **A Sequential Multi-Classification Method for sMRI Images**, invention patent 2024.08
  Granted, CN116740463B; fifth inventor. A ranking model keeps distances between class feature vectors consistent with their distances along the actual pathological progression. Rather than treating the classes as mutually independent labels, the design carries the ordinal structure of disease progression into the feature space, improving multi-class performance.
- **sMRI Classification with a High-Resolution Complementary-Attention UNet**, invention patent 2022.07
  Granted, CN114332535B; fifth inventor. Feature-fusion and classification branches inserted into UNet supply semantic information to the encoder and detail to the decoder. The classification branch fuses multi-semantic feature maps to raise the network's representational capacity and, with it, classification accuracy.
- **Fanxing Sanyi Information Consulting Mini Program**, software copyright 2023.05
  A consulting Mini Program for Zhejiang's "Trinity" admission scheme: it recommends eligible universities from a candidate's academic exam results and provides per-university strategies and admission records. I was responsible for the whole build, with storage on WeChat Cloud's database.

Awards and Honours

- Outstanding Undergraduate Student, Ningbo University 2022–2023
- Outstanding Student Cadre, Ningbo University 2021–2022

Technical Skills

- **Programming**: Python (proficient), C/C++, SQL, JavaScript/HTML/CSS, LaTeX
- **Machine learning**: PyTorch, HuggingFace Transformers / PEFT (LoRA) / TRL (DPO), scikit-learn, NumPy/SciPy, YOLO, multi-GPU training
- **Tools and platforms**: Linux, macOS, Git, MongoDB, openGauss, WordPress
- **Languages**: Chinese (native), English (MSc taught in English; research written in English)